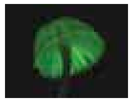
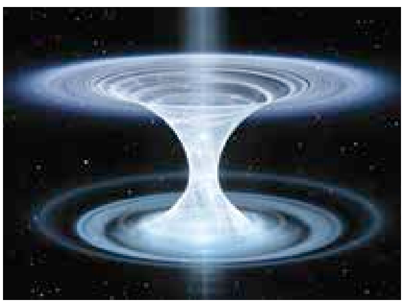
Former Batman no more
Yesteryear's Batman of the US TV series in the 1960s, Adam West passes away at the age of 88 due to leukaemia. **http://dgit.in/BatOut**



Glow in the dark shrooms
Certain mushrooms are able to emit light due to an enzyme pre-maturely re-jiggering a compound called hispidin. **http://dgit.in/shroomzz**

negative energy within the wormhole in order to keep its walls from collapsing, killing our time traveller in the process. This means that first we have to extract all energy and create pure vacuum, and then suck some more energy out of the portal. It's like creating an energy debt, but there are limitations on how much energy we can extract from a region that already has none, at least according to classical physics.

In his paper on Space and Time Warps, Stephen Hawking explains how it may be possible to build a wormhole, "Energy is rather like money. If you have a positive bank balance, you can distribute it in various ways. But according to the classical laws that were believed until quite recently, you weren't allowed to have an energy overdraft. So these classical laws would have ruled out us being able to warp the universe, in the way required to allow time travel. However, the classical laws were overthrown by Quantum Theory, which is the other great revolution in our picture of the universe, apart from General Relativity. Quantum Theory is more relaxed, and allows you to have an overdraft on one or



If black holes bore you, there are also these funnel-shaped portals called wormholes, though we haven't spotted one in nature yet, these will most likely kill you too

two accounts. If only the banks were as accommodating. In other words, Quantum Theory allows the energy density to be negative in some places, provided it is positive in others."

Let's go back to the spaceship, and see if we can get it to travel faster than light. In 1994, Miguel Alcubierre devised a model called Alcubierre drive which involves warping the space in and around the ship to arrive at destinations faster than light without depending on your own kinetic energy, a bit like surfing a wave. Alas, this is the most speculative of concepts that we've discussed so far. NASA was working on it at some point but have since stopped. A statement released by them says, "Science fiction writers have given us many images of interstellar travel, but travelling at the speed of light is simply imaginary at present. In the meantime, science moves forward. And while NASA is not pursuing interstellar flight, scientists here continue to advance ion propulsion for missions to deep space and beyond using solar electric power. This form of propulsion is the fastest and most efficient to date. There are many "absurd" theories that have become reality over the years of scientific research. But for the near future, warp drive remains a dream."

PARADOXES

Let's say you manage to get your hands on a sweet time machine, and also, you hate your grandfather for reasons best known to you. You get in the machine, dial back a few decades, and go and kill your grandfather before he fathered your father. Now wait a second, if your grandfather died before your father was conceived, how did you exist to be able to come back and kill your own grandpa? This conundrum is aptly called the Grandfather Paradox, and it's just one of the many problems we encounter when thinking about time travel.

Opinion is divided on the matter, some physicists say that an yet undiscovered physical law will stop us from changing the past. Another, more interesting explanation says that an alternate reality will branch off when you kill your grandfather, you don't exist in that reality but that doesn't stop you from existing in the one you came from. This many-worlds interpretation suggests the existence of a multiverse where anything that can happen, does happen.

Sometimes, objects and information can get stuck in time. This is known as the Bootstrap Paradox. Suppose a time traveller comes back in time and explains relativity to Einstein, who then builds a theory around the idea—his theory gains popularity and soon becomes one of the most discussed scientific concepts in the history of mankind. Then, in the future, the time traveller who came back, learns about it in school. Now, who really created the theory?

According to the many-worlds interpretation, our universe may be just one of the countless other universes and alternate realities that exist, in case you were feeling important

Clearly not Einstein, because the time traveller informed him about it, but also not the time traveller, who didn't know about it until he learned it from a textbook. In the movie Somewhere in Time, an old woman gives Christopher Reeves a pocket watch in 1972, later Reeves travels back to the year 1912 and gives the watch back to the woman who is now a young girl. Where did the watch really originate? In both these cases, the information and object are trapped within an infinite loop of cause and effect, with no clear origin outside of that loop.

Time travel is one of those big unsolved scientific questions that forces us to think deeply, and then discover and invent things to turn our vision into a reality. Obviously, barring the exception of stumbling upon a functional time machine, it's unlikely that we're going to crack the code anytime in the near future, but some day in the distant future, taking a trip in time may be as easy as calling a cab to go downtown. Till then, you can always just catch another flight to arrive a few nanoseconds into the future.